



Volumes of Solids of Revolution: The Washer Method

Calculus Worksheet · Grade 11-12

Name: _____

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Learning Objectives

- Identify outer and inner radii for solids of revolution with a hole in the middle
- Set up definite integrals using the washer method formula
- Compute volumes of solids generated by rotating regions about horizontal and vertical axes

For each problem, sketch the region, identify the outer and inner radii, set up the washer-method integral, and evaluate to find the exact volume.

1. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^1 (x^2 - (x^2)^2) dx$$

Answer: _____

2. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the line $y = -1$.

$$V = \pi \int_0^1 ((x+1)^2 - (x^2+1)^2) dx$$

Answer: _____

3. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^1 (x - x^4) dx$$

Answer: _____

4. Find the volume of the solid generated by revolving the region bounded by $y = 2x$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^2 (4x^2 - x^4) dx$$

Answer: _____

5. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 4$ about the x-axis.

$$V = \pi \int_{-2}^2 (16 - x^4) dx$$

Answer: _____



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Volumes of Revolution — Washer Method — Set B Worksheet

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6. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the line $y = 2$.

$$V = \pi \int_0^1 ((2 - x^2)^2 - (2 - x)^2) dx$$

Answer: _____

7. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 2x$ about the line $y = 5$.

$$V = \pi \int_0^2 ((5 - x^2)^2 - (5 - 2x)^2) dx$$

Answer: _____

8. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = x$ about the line $y = -2$.

$$V = \pi \int_0^1 ((x + 2)^2 - (x^2 + 2)^2) dx$$

Answer: _____

9. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ about the line $y = 3$.

$$V = \pi \int_0^4 (9 - (3 - \sqrt{x})^2) dx$$

Answer: _____

10. Find the volume of the solid generated by revolving the region bounded by $y = x^3$ and $y = x$ about the x -axis in the first quadrant.

$$V = \pi \int_0^1 (x^2 - x^6) dx$$

Answer: _____





Encourage students to always sketch the region and the axis of rotation before writing the integral; emphasize that the washer formula requires squaring each radius separately, not subtracting first.

Solutions

1. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^1 (x^2 - (x^2)^2) dx$$

- Find the intersection points by setting x equal to x squared, giving x equals 0 and x equals 1.
- The outer radius is the line y equals x and the inner radius is the parabola y equals x squared.
- Set up the washer integral with π times the integral from 0 to 1 of x squared minus x to the fourth dx .
- Integrate term by term to get x cubed over 3 minus x to the fifth over 5 evaluated from 0 to 1.
- Substitute the limits to get one third minus one fifth, which equals two fifteenths.
- Multiply by π to obtain the final volume of two π over fifteen.

Answer: $V = \frac{2\pi}{15}$

2. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the line $y = -1$.

$$V = \pi \int_0^1 ((x+1)^2 - (x^2+1)^2) dx$$

- Shift both functions up by 1 to account for the axis of rotation at y equals negative 1.
- The outer radius is x plus 1 and the inner radius is x squared plus 1.
- Expand the squares to get x squared plus $2x$ plus 1 minus x to the fourth plus $2x$ squared plus 1.
- Simplify the integrand to $2x$ minus x squared minus x to the fourth.
- Integrate from 0 to 1 to get x squared minus x cubed over 3 minus x to the fifth over 5.
- Evaluate to get 1 minus one third minus one fifth, which equals twenty-nine over thirty, then multiply by π .

Answer: $V = \frac{29\pi}{30}$

3. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^1 (x - x^4) dx$$

- Find the intersection points by setting square root of x equal to x squared, giving x equals 0 and x equals 1.
- The outer radius is square root of x and the inner radius is x squared.
- Square each radius to obtain x and x to the fourth respectively.
- Set up the integral π times the integral from 0 to 1 of x minus x to the fourth dx .
- Integrate to get x squared over 2 minus x to the fifth over 5 evaluated from 0 to 1.
- Compute one half minus one fifth to obtain three tenths, then multiply by π .

Answer: $V = \frac{3\pi}{10}$



4. Find the volume of the solid generated by revolving the region bounded by $y = 2x$ and $y = x^2$ about the x-axis.

$$V = \pi \int_0^2 (4x^2 - x^4) dx$$

- Find intersections by setting $2x$ equal to x squared, giving x equals 0 and x equals 2.
- The outer radius is $2x$ and the inner radius is x squared.
- Square the radii to get $4x$ squared and x to the fourth.
- Integrate π times the integral from 0 to 2 of $4x$ squared minus x to the fourth dx .
- Evaluate the antiderivative $\frac{4x^3}{3}$ minus $\frac{x^5}{5}$ at the bounds.
- Simplify to $\frac{32}{3}$ minus $\frac{32}{5}$, giving $\frac{64}{15}$ times π .

Answer: $V = \frac{64\pi}{15}$

5. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 4$ about the x-axis.

$$V = \pi \int_{-2}^2 (16 - x^4) dx$$

- Determine the bounds by setting x squared equal to 4, giving x equals negative 2 and x equals 2.
- The outer radius is the constant 4 and the inner radius is x squared.
- Square each radius to get 16 and x to the fourth.
- Integrate π times the integral from negative 2 to 2 of 16 minus x to the fourth dx .
- Compute the antiderivative $16x$ minus $\frac{x^5}{5}$ at the limits.
- Simplify to get $\frac{256\pi}{5}$ as the final volume.

Answer: $V = \frac{256\pi}{5}$

6. Find the volume of the solid generated by revolving the region bounded by $y = x$ and $y = x^2$ about the line $y = 2$.

$$V = \pi \int_0^1 ((2 - x^2)^2 - (2 - x)^2) dx$$

- Since the axis $y = 2$ is above the region, the outer radius is 2 minus x squared and the inner radius is 2 minus x .
- Expand each squared expression carefully to get 4 minus $4x$ squared plus x to the fourth minus the expansion of 2 minus x squared.
- Simplify the integrand to x to the fourth minus $4x$ squared minus x squared plus $4x$, then combine like terms.
- Integrate the simplified polynomial from 0 to 1.
- Evaluate the antiderivative at the bounds.
- Multiply by π to obtain $\frac{8\pi}{15}$.

Answer: $V = \frac{8\pi}{15}$

7. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = 2x$ about the line $y = 5$.

$$V = \pi \int_0^2 ((5 - x^2)^2 - (5 - 2x)^2) dx$$

- Determine intersection points where x squared equals $2x$, giving x equals 0 and x equals 2.
- Because $y = 5$ lies above the region, outer radius is 5 minus x squared and inner radius is 5 minus $2x$.
- Expand both squared terms and subtract carefully.
- Combine like terms to obtain a polynomial integrand.
- Integrate from 0 to 2 and evaluate at the bounds.
- Multiply the result by π to get $\frac{136\pi}{15}$.

Answer: $V = \frac{136\pi}{15}$



8. Find the volume of the solid generated by revolving the region bounded by $y = x^2$ and $y = x$ about the line $y = -2$.

$$V = \pi \int_0^1 ((x+2)^2 - (x^2+2)^2) dx$$

- Shift each function by adding 2 since the axis of rotation is y equals negative 2.
- The outer radius is x plus 2 and the inner radius is x squared plus 2.
- Expand each squared radius to get x squared plus $4x$ plus 4 and x to the fourth plus $4x$ squared plus 4.
- Subtract to obtain the integrand $4x$ minus $3x$ squared minus x to the fourth.
- Integrate from 0 to 1 to get $2x$ squared minus x cubed minus x to the fifth over 5.
- Evaluate to obtain seventeen fifteenths and multiply by π .

Answer: $V = \frac{17\pi}{15}$

9. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ about the line $y = 3$.

$$V = \pi \int_0^4 (9 - (3 - \sqrt{x})^2) dx$$

- Identify the outer radius as 3 (distance from x -axis to y equals 3) and the inner radius as 3 minus square root of x .
- Square the radii to get 9 and 9 minus 6 square root of x plus x .
- Subtract to simplify the integrand to 6 square root of x minus x .
- Integrate from 0 to 4, giving $4x$ raised to the three halves minus x squared over 2.
- Evaluate the antiderivative at x equals 4.
- Multiply by π to obtain forty π over three.

Answer: $V = \frac{40\pi}{3}$

10. Find the volume of the solid generated by revolving the region bounded by $y = x^3$ and $y = x$ about the x -axis in the first quadrant.

$$V = \pi \int_0^1 (x^2 - x^6) dx$$

- Find intersections in the first quadrant by setting x cubed equal to x , giving x equals 0 and x equals 1.
- Outer radius is the line y equals x and inner radius is the curve y equals x cubed.
- Square the radii to obtain x squared and x to the sixth.
- Set up the integral π times the integral from 0 to 1 of x squared minus x to the sixth dx .
- Integrate to get x cubed over 3 minus x to the seventh over 7.
- Evaluate at the bounds to find one third minus one seventh, equaling four twenty-firsts, and multiply by π .

Answer: $V = \frac{4\pi}{21}$

